

PROJECT #2 SAMUEL SÁNCHEZ GARCÍA

Problem 1)

Design a Dolph-Tschebyscheff broadside array of 8 elements, 20-dB sidelobe level, and with a spacing of $\lambda/4$ between the elements to operate at 4 GHz. Choose a “rectangular patch” or a "dipole" antenna as the array element that operates at 4 GHz. For the final antenna design, determine the:

- Radiation pattern
- Gain in dB
- Half-power beamwidth (*in degrees*). How much narrower or wider (*in degrees*) is this half-power beamwidth compared to that of a uniform array of the same number of elements and spacing?
- How much smaller or larger (*in dB*) is the gain (*in dB*) compared to that of a uniform array of the same number of elements and spacing? *Justify your answer. Do you expect it? Why?*

Problem 1: Dolph-Tschebyscheff Broadside Array

For this design, I selected half-wavelength dipole $\lambda/2$ operating at a frequency of 4 GHz to model at CST. At this frequency, the free-space wavelength is $\lambda = 75$ mm. The physical spacing between the array elements is $d = \lambda/4 = 18.75$ mm.

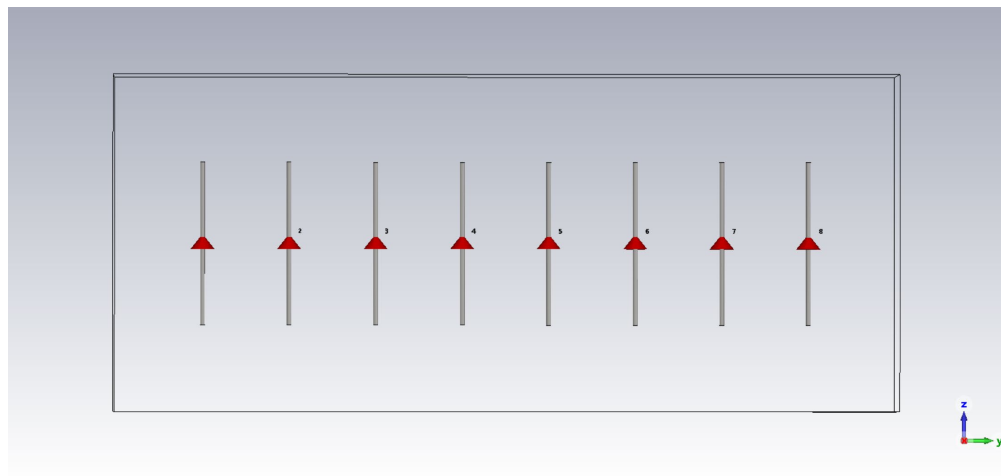


Figure 1.1: Simulation CAD model of the array of dipoles.

The design requires an array of $N = 8$ elements with a sidelobe level (SLL) of -20 dB.

- **Voltage Ratio (R):** A -20 dB SLL corresponds to a voltage ratio of $R = 10$, since $SLL = 20 \log_{10}(R)$.
- **Chebyshev Polynomial:** For 8 elements, the pattern is governed by a Chebyshev polynomial of degree $m = N - 1 = 7$, which is $T_7(x)$.
- **Scaling Factor (x_0):** This is calculated by equating the polynomial to the voltage ratio, $T_7(x_0) = 10$. The result is $x_0 = \cosh [1/7 \cosh^{-1}(10)] \approx 1.0927$.
- **Excitation Coefficients:** The array factor (AF) for an 8-element symmetric array is expanded using trigonometric identities into powers of z and equated to $T_7(z \cdot z_0)$:

$$z^7(64a_4) + z^5(16a_3 - 112a_4) + z^3(4a_2 - 20a_3 + 56a_4) + z(a_1 - 3a_2 + 5a_3 - 7a_4) = 64(z_0z)^7 - 112(z_0z)^5 + 56(z_0z)^3 - 7(z_0z)$$

Matching the coefficients for each power of z and normalizing relative to the outer edge elements gives the final amplitudes:

- $I_1, I_8 = 1.000$
- $I_2, I_7 = 1.137$
- $I_3, I_6 = 1.506$
- $I_4, I_5 = 1.720$

Final Antenna Design Determinations

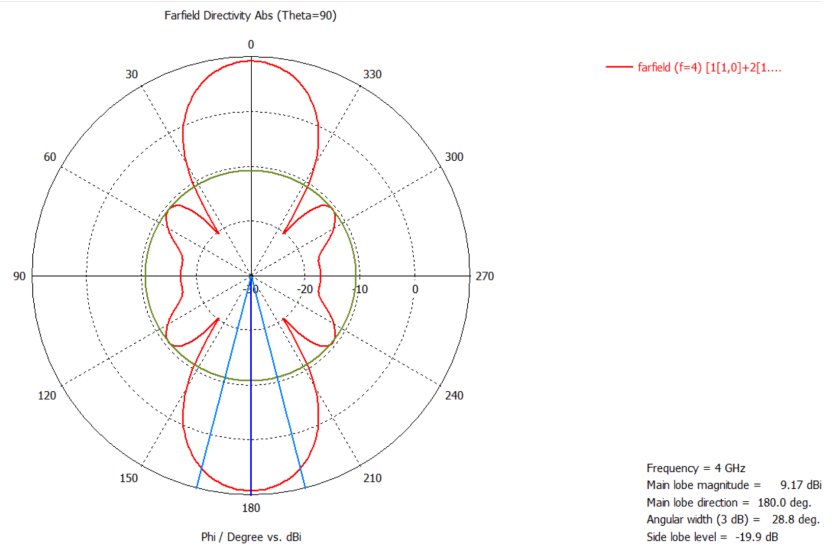


Figure 1.2: 1D Polar cut of the array's radiation pattern at ($\theta = 90^\circ$).

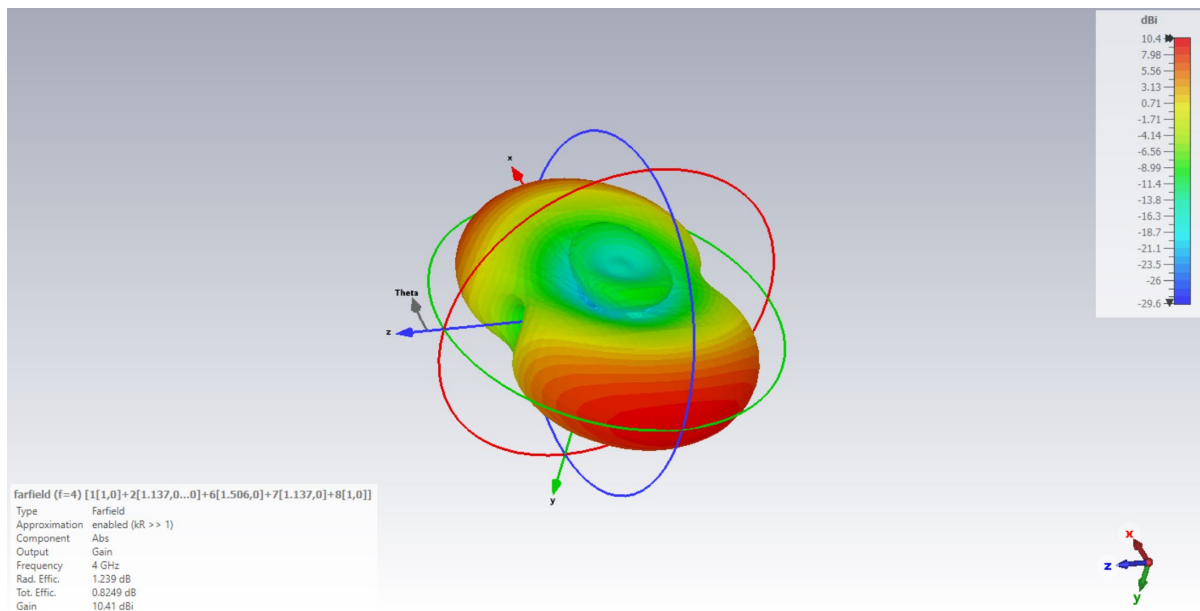


Figure 1.3: 3D Radiation Pattern of the Uniform Array

- **Radiation Pattern:** The resulting radiation pattern shows the main beam pointing in the broadside direction with the maximum side lobe level rigidly set at roughly -20 dB.
- **Gain in dBi:** The total gain of the array is **10.4 dBi**.
- **Half-power beamwidth (in degrees):** The half-power beamwidth is **28.8 degrees**.

4. Comparison with a Uniform Array To perform the comparison, the same array was evaluated by applying a uniform amplitude excitation ($I_n = 1$ for all elements).

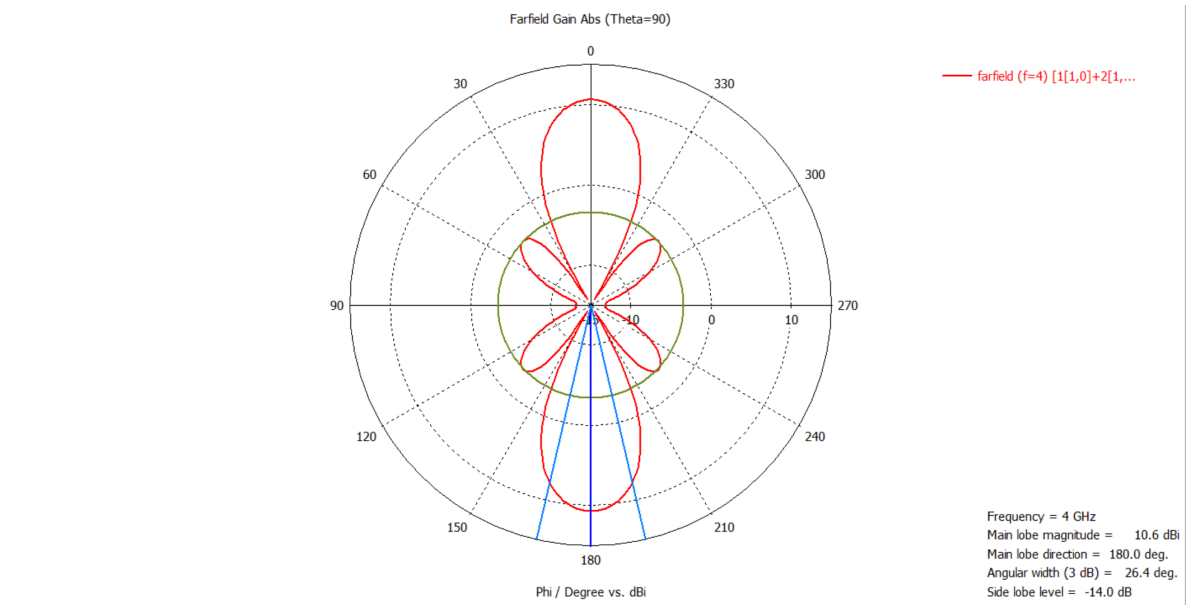


Figure 1.4: 1D Polar cut of the array's radiation pattern at ($\theta = 90^\circ$).

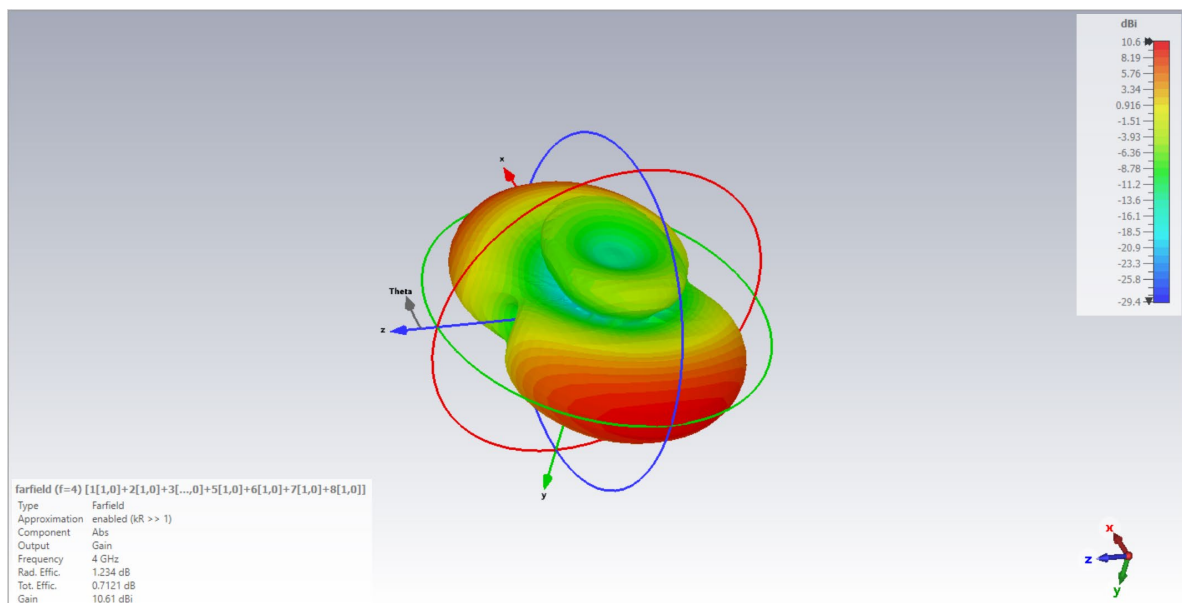


Figure 1.5: 3D Radiation Pattern of the Uniform Array

- **How much narrower or wider?**

The half-power beamwidth (HPBW) of the Dolph-Tschebyscheff array is **wider** in comparison to the uniform array. Specifically, it has widened by **2.4 degrees**.

- **How much smaller or larger is the gain?**

The gain of the Dolph-Tschebyscheff array is **smaller** than that of the uniform array. Specifically, the gain has been reduced by **0.2 dB**.

5. Justification: Do you expect it? Why?

Yes, this is completely expected. A uniform array distribution always provides the absolute maximum directivity, the highest gain, and the narrowest main beam for a given aperture size, since its aperture illumination efficiency is 100%. However, its side lobes are naturally very high.

To suppress the side lobes down to -20 dB, the Dolph-Tschebyscheff method is forced to apply a strong amplitude taper across the array aperture, exciting the central elements with much more power than the edge elements. Generally, in antenna engineering, applying this non-uniform tapering reduces the effective aperture efficiency, which fundamentally forces a sacrifice in total gain and a widening of the main beam.

Problem 2)

Design a 3 x 4 element uniform planar array (3 elements along x and 4 along y) at 6 GHz. If the spacing of $dx = \lambda/2$, and $dy = \lambda/4$ between the elements, find the:

- progressive phase shift between the elements in the x and y directions that will give a main beam at $\theta = 30^\circ$ and $\phi = 30^\circ$. Once you calculate these values choose a $\lambda/2$ dipole element to determine:
- the total gain of the array, and
- the total Radiation Pattern Calculations performed.

Problem 2: Design and Simulation of a 3x4 Planar Array at 6 GHz

Theoretical Calculations

The objective of this design is to characterize a uniform 3x4 planar array operating at $f = 6$ GHz. At this frequency, the free-space wavelength is $\lambda = c/f = 50$ mm. The geometric spacing between the elements is defined as $d_x = \lambda/2 = 25$ mm and $d_y = \lambda/4 = 12.5$ mm.

To scan the main beam toward the specific angular coordinates of $\theta_0 = 30^\circ$ and $\phi_0 = 30^\circ$, a progressive phase shift must be applied across the elements in both the x and y directions. By setting the total phase Ψ to zero in the desired direction of maximum radiation, the phase shifts α_x and α_y are calculated as follows:

For the x-direction ($d_x = \lambda/2$):

$$\alpha_x = -(2\pi/\lambda) * d_x * \sin(\theta_0) * \cos(\phi_0) = -\pi * 0.5 * \sin(30^\circ) * \cos(30^\circ) = -77.94^\circ$$

For the y-direction ($d_y = \lambda/4$):

$$\alpha_y = -(2\pi/\lambda) * d_y * \sin(\theta_0) * \sin(\phi_0) = -(\pi/2) * \sin(30^\circ) * \sin(30^\circ) = -22.5^\circ$$

According to the Principle of Pattern Multiplication, the total radiation pattern of an array composed of identical elements is the mathematical product of the isolated single-element pattern and the normalized Array Factor (AF). Choosing a $\lambda/2$ dipole oriented along the Y-axis as the radiating element, the total electric field pattern is expressed as:

$$E_{\text{total}}(\theta, \varphi) = E_{\text{element}}(\theta, \varphi) * AF_n(\theta, \varphi)$$

Where the normalized Array Factor for the 3x4 planar array is:

$$AF_n(\theta, \varphi) = \left[\frac{1}{3} * \frac{\sin(1.5 * \Psi_x)}{\sin(0.5 * \Psi_x)} \right] * \left[\frac{1}{4} * \frac{\sin(2 * \Psi_y)}{\sin(0.5 * \Psi_y)} \right]$$

(with $\Psi_x = \pi * \sin(\theta) * \cos(\varphi) - 77.94^\circ$ and $\Psi_y = (\pi/2) * \sin(\theta) * \sin(\varphi) - 22.5^\circ$)

Assuming a lossless array, the directivity of the Array Factor for a planar array scanned near broadside can be approximated using the broadside directivities for linear arrays ($D_x \approx 2N(d_x/\lambda)$ and $D_y \approx 2N(d_y/\lambda)$):

$$D_{\text{AF}} \approx \pi * \cos(\theta_0) * D_x * D_y = \pi * \cos(30^\circ) * 3 * 2 \approx 16.32 \text{ (or 12.13 dB)}$$

Combining this with the peak directivity of a half-wave dipole (2.15 dBi), the theoretical maximum expected gain is approximately 14.28 dBi.

Electromagnetic Simulation and Results

To prevent the severe gain drop-off that a Z-oriented dipole exhibits at $\theta = 30^\circ$ (mistake I made), the $\lambda/2$ dipole (length = 25 mm) was modeled parallel to the Y-axis.

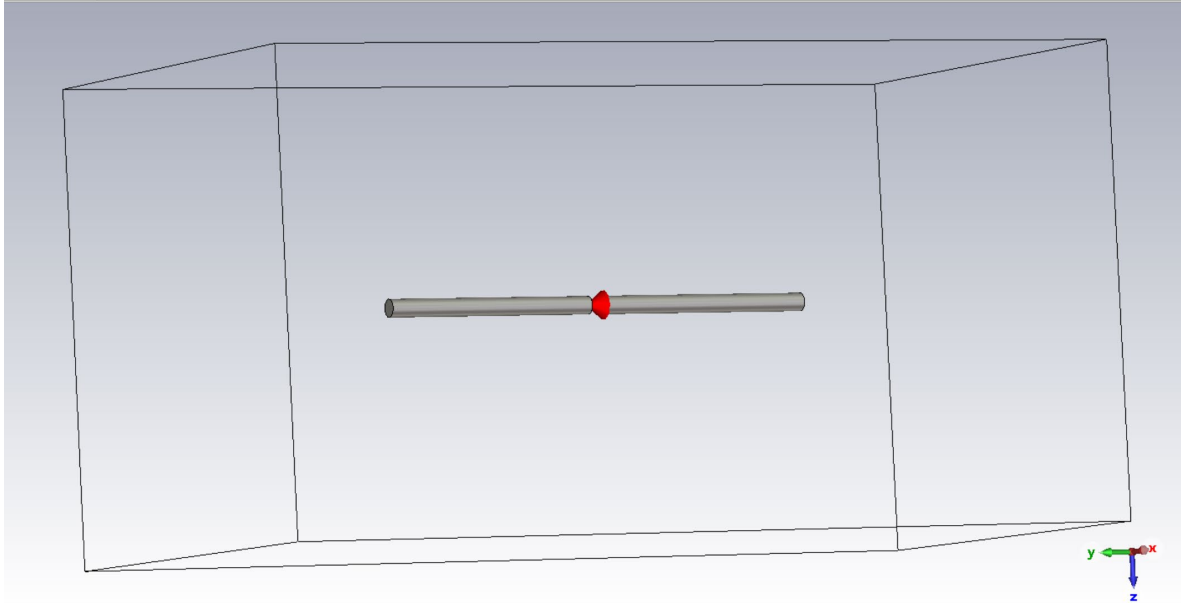


Figure 2.1: Simulation CAD model of the isolated dipole oriented along the Y-axis at 6 GHz.

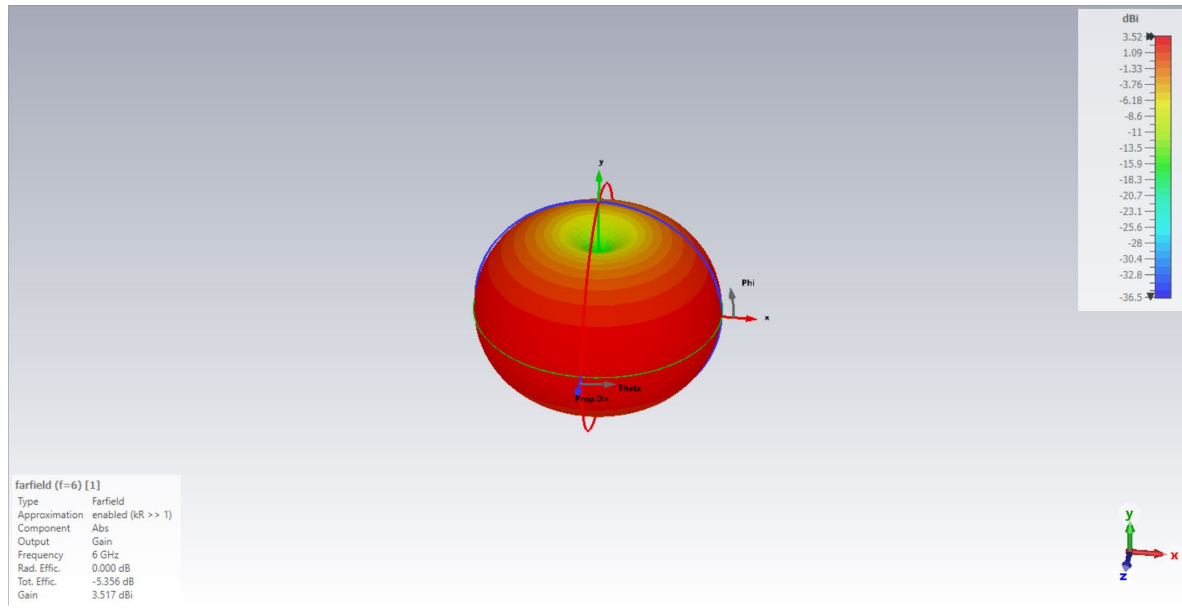


Figure 2.2: 3D Radiation Pattern of the isolated Y-oriented dipole at 6 GHz, simulated in CST Microwave Studio.

After simulating the single element, the previously calculated progressive phase shifts ($\alpha_x = -77.94^\circ$, $\alpha_y = -22.5^\circ$) and spacings ($d_x = 25$ mm, $d_y = 12.5$ mm) were introduced into the array configuration matrix to synthesize the 3x4 array.

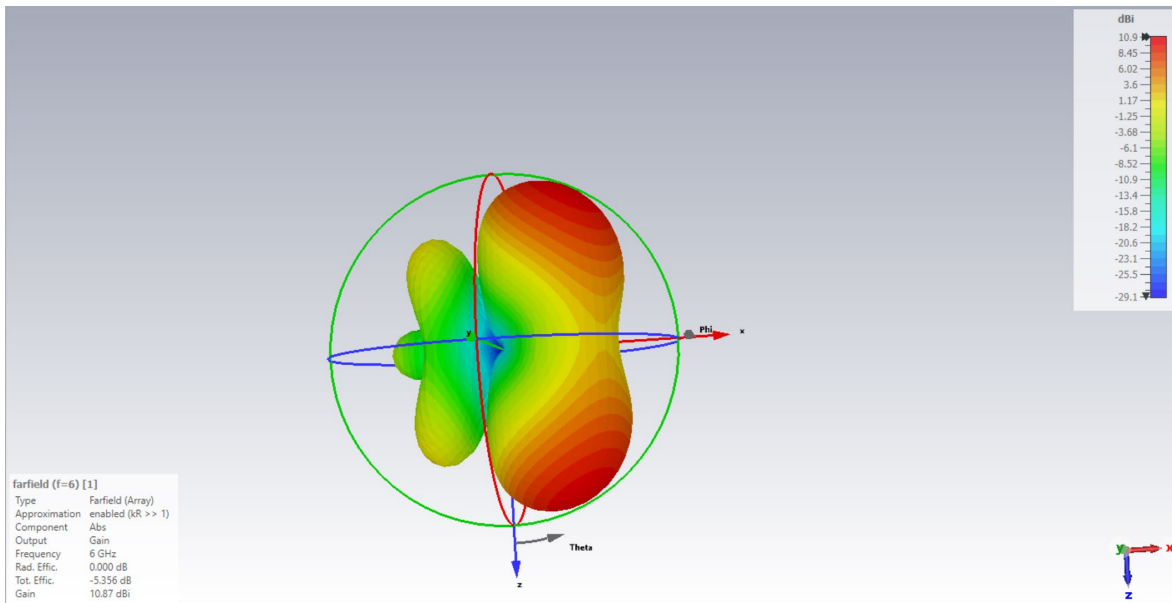


Figure 2.3: 3D Radiation Pattern of the 3x4 planar array demonstrating the steered main beam.

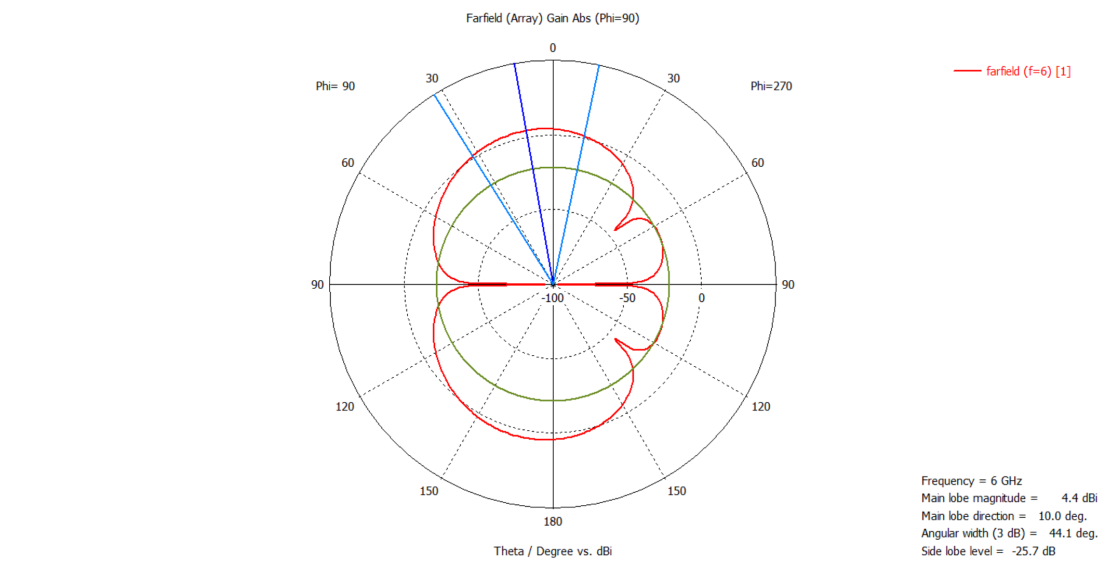


Figure 2.4: 1D Polar cut of the array's radiation pattern at ($\phi=90^\circ$), illustrating the peak gain at ($\theta = 28^\circ$).

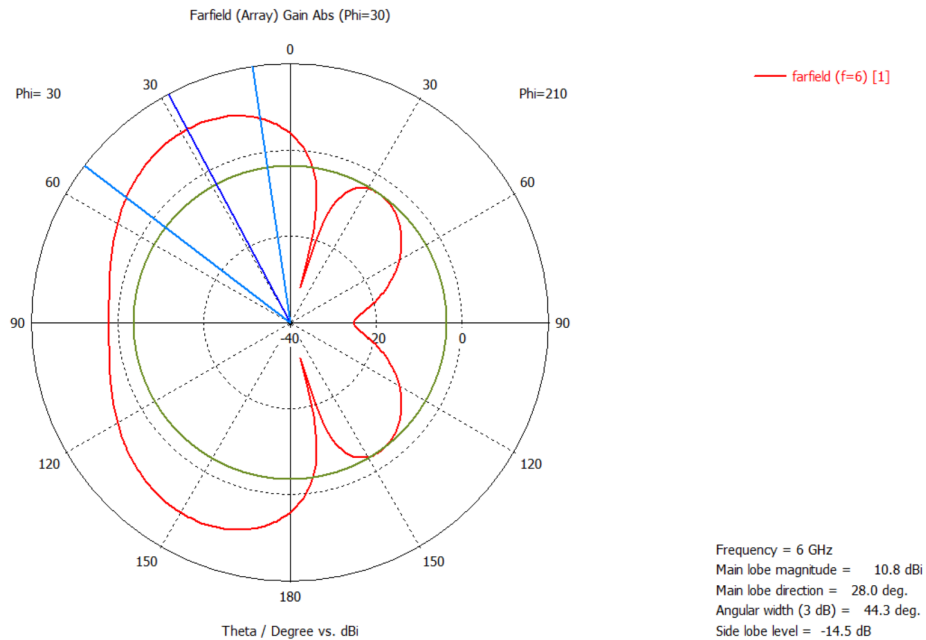


Figure 2.5: 1D Polar cut of the array's radiation pattern in the YZ-plane ($\phi=90^\circ$), illustrating the off-axis peak at ($\theta = 10^\circ$).

The exact numerical results extracted from the CST simulation were:

Total Realized Gain: **10.8 dB**.

Main Lobe Direction (Scanning Plane $\phi=30^\circ$): **$\theta=28.0^\circ$** .

Main Lobe Direction (Orthogonal Plane $\phi=90^\circ$): **$\theta=10.0^\circ$** .

Conclusions and Physical Analysis

The implementation of the theoretical parameters into a full-wave electromagnetic simulator yielded numerical results that slightly diverge from the ideal mathematical approximations. These discrepancies are not software errors, but rather fundamental electromagnetic phenomena:

Gain Overestimation in Analytical Formulas

The theoretical gain of 14.28 dBi is heavily overestimated. The theoretical formula ($D_{AF} \approx \pi * \cos(\theta_0) * D_x * D_y$) is an asymptotic approximation valid strictly for electrically large planar arrays. For a small array of only 12 elements, the absolute maximum directivity for independent isotropic elements is $10 * \log_{10}(12) \approx 10.79$ dB. Therefore, the 10.9 dB gain provided by CST is the rigorously correct value, obtained through exact 3D spherical numerical integration rather than a simplified approximation.

Inward Beam Squint in the Main Scanning Plane ($\varphi=30^\circ$)

While the mathematical Array Factor was perfectly phased to point to its maximum at exactly $\theta=30^\circ$, the simulated main lobe shifted inward to $\theta=28.0^\circ$. This occurs precisely due to the Pattern Multiplication Principle. The single dipole oriented along the Y-axis has its maximum radiation intensity at broadside ($\theta=0^\circ$), and its pattern naturally decays as θ increases. When multiplying this sloped element pattern by the Array Factor's peak at 30° , the higher relative intensity of the dipole at 28° pulls the final resulting maximum inward toward a new equilibrium angle. This phenomenon is a textbook demonstration of Beam Squint.

Off-Axis Beam Squint in the Elevation Plane ($\varphi=90^\circ$)

In the $\varphi=90^\circ$ plane, the array factor phase term in the X-direction becomes constant ($\cos(90^\circ) = 0$). The pattern shape is governed strictly by the Y-direction phase term:

$$\Psi_y = \beta * d_y * \sin(\theta) * \sin(90^\circ) + \alpha_y = 90^\circ * \sin(\theta) - 22.5^\circ$$

Equating $\Psi_y = 0$ yields a theoretical maximum for the Array Factor at $\theta \approx 14.47^\circ$. However, because the dipole is physically oriented along the Y-axis, the $\varphi=90^\circ$ cut corresponds to the E-plane of the radiating element, which peaks strictly at broadside ($\theta=0^\circ$). The decreasing slope of the dipole's element pattern from 0° pulls the theoretical array factor peak

downwards, yielding the exact mathematical product equilibrium point at $\theta=10.0^\circ$, perfectly matching the CST simulation.

Constructive and Physical Incompatibilities

A critical physical limitation is revealed by the geometry of this exercise. At 6 GHz, a $\lambda/2$ dipole measures 25 mm in length. Since the elements are oriented along the Y-axis, and the required spacing along that same Y-axis is $d_y = \lambda/4 = 12.5$ mm, the physical antennas would severely overlap and intersect each other in reality. This confirms that Problem 2 is a purely theoretical exercise designed exclusively to validate the Pattern Multiplication Principle, which was successfully demonstrated here using CST's post-processing capabilities without triggering physical collisions.